Weighted Dilation Graphs and Totient Fibers: Ram Energy Representations and Spectrally Truncated Ideal Kernels

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M021

October 8, 2026

Abstract

We examine the weighted dilation-graph and independent-label stages of the September 24, 2026 manuscript on smooth shifted primes and totient fibers. The physical operator is generally not self-adjoint; its state measure, adjoint, fiber-constant root vectors, and positive singular-value energies are retained explicitly. The principal local result replaces the source's conditional character frame by a spectrally truncated projection. This gives norm-one log-cell kernels, controlled coefficient complexity, and an explicit comparison error without near-orthogonality or a lower Gram eigenvalue hypothesis. In this comparison step the prime groups need not be intervals. Further proved refinements reduce the probe-error sum from 3^P to less than $5^{P/2}$, reduce the sufficient memory cutoff, and weaken the endpoint amplitude assumption to a local coefficient-energy condition. Exact finite Fourier, Mellin, and Gram descriptions identify what these changes do and do not establish. Shifted arithmetic correlations, Type II savings, prime mass, and the global totient conclusion are not certified by this report. Contribution assessment: C3/5, a proved weaker-assumption ideal-kernel comparison and quantitative local refinements leave the decisive arithmetic and extraction obligations open.

Contents

1 Scope, architecture, and evidence **2**

2 The two source operators and their measures **3**

3 Frequency and logarithmic representations **4**

4 A spectrally truncated replacement for ideal cells **5**

5 Sharper symmetric-probe accounting **8**

6 Memory cutoff and endpoint assumptions **10**

7 Finite certificates and their failure modes **11**

8 Arithmetic obligations and the totient boundary **12**

9 Public comparison, costs, and research priorities **12**

10 Verification and contribution assessment **13**

1 Scope, architecture, and evidence

The assigned source is OpenAI's *Weighted dilation graphs, smooth shifted primes and totient fibers* [1], catalogue entry 21, applicability L3. Its claimed conclusions are, for each fixed $\delta > 0$, $x^{1-o(1)}$ primes $2x < p \le 5x$ with $P^+(p-1) \le x^\delta$, and, for each $\varepsilon > 0$, infinitely many n with $\#\{m : \varphi(m) = n\} > n^{1-\varepsilon}$. These are *source claims*, not theorems independently established here. Applicability and contribution are different ratings; L3 is preserved regardless of the C3 assessment.

The parent authenticated the source subtree as unchanged between the pinned base and the captured October 8 target. This is byte provenance, not proof validity, and is not an independent live-branch authentication by this author. All ten assigned TeX files were hashed before scientific reading. Selected private originals and the bibliography/README were also hashed before their reads. Final matching hashes certify endpoint equality, not continuous monitoring.

Here and below `02`, for example, means `sections/02-transference.tex` in the assigned build directory. The complete section architecture was mapped. The following is the actual literal-read scope, not a claim of full-proof verification.

Source unit	Lines literally read	Function and remaining boundary
`paper`, preamble	1–27; 1–57	Includes, conventions, author and claims.
`00` introduction	1–245	Claims and complete dependency narrative.
`01` preliminaries	1–85	Scales and choice order; later sieve and analytic estimates imported.
`02` transference	1–1339, bounded chunks	Physical measures, memory, rare transfers, clean Fourier reduction, equality rank, pairing. Read in full, not globally certified.
`03` ideal kernels	1–645, bounded chunks	Product atoms, bilinear estimate, cells, comparison, probe cancellation. Main local target.
`04` correlations	1–155; 421–536	Endpoint hypotheses and local Mellin energy. Remaining shift proof uninspected.
`05` Type II	1–118; 337–431; 512–561	Statement, determinant map and closing dependencies; intervening estimates imported.
`06` prime extraction	1–102; 869–922	Candidate and final parameter order; mass, proxy, sieve and balanced-range proofs mostly uninspected.
`07` totients	1–90	Conditional product/pigeonhole deduction read. Prime supply not established here.
Bibliography; README	1–245; 1–18	Bibliographic routing and public identification, not cited proofs.

The scientific route is

$$(\text{physical transference} + \text{ideal kernels}) \longrightarrow \text{shifted correlations} \longrightarrow \text{Type II},$$

followed by candidate mass, Type I distribution, rough proxies and two composite-removal regimes, then smooth primes and totient fibers. Improving the ideal-kernel branch does not verify these other branches.

Private text actually read comprises TR190 lines 1–318, volume 2, chapter 1 lines 447–543, and TR1674 revised v2 lines 112–380 [6, 7, 8]. TR190 supplies finite spectral calculus and polynomial locality, not an arithmetic norm bound. The chapter gives the actual Mellin inversion/Plancherel proof. TR1674 gives the actual energy/correlation proof, quarter-scale algebra and finite Gram formulation; its external RH sources and broader private corpus were not read here. Neither its Liouville cancellation nor its quarter-scale factor is transferred to this graph. Only the exact energy methodology is used.

2 The two source operators and their measures

Put $L = \log x$, $T = \log L$. All constants are fixed before $x \to \infty$. The disjoint small and big prime groups have $s \asymp T$ groups in total, with primes in $[\exp(L^a), \exp(L^b)]$ and $[\exp(L^c), \exp(L^d)]$, respectively, where $0 < a < b < c < d < 0.47$. Write

$$V_g = \sum_{p \in \mathcal{P}_g} p^{-1} \in [v_-, v_+], \qquad \mu_g(p) = (V_g p)^{-1}, \qquad M = J + \ell,$$

where $0 < q < 1$, $\ell \geq 1$, and J is a large fixed integer. An ideal big-group list has M independent slots of law μ_g; repetitions are allowed. A physical list has M distinct primes per group, all dividing its integer position n. Its state mass is

$$d\mu_{\mathrm{ph}}(n, \mathbf{p}) = \prod_g V_g^{-M} \quad \text{per admissible ordered state,} \tag{2.1}$$

with counting measure in $n \in \mathbb{Z}$. It is not a probability at each n.

A pattern frees $t_g = \ell + |C_g|$ slots, where $|C_g| \leq m$ in big groups and $C_g = \varnothing$ in small groups. Shared labels are retained; auxiliary labels fill the freed positions in the Js-factor product D. The physical row moves to $n' = n + kD$, $0 < |k| \leq L^{C_2}$, sums admissible target lists with coefficient $\prod_g V_g^{-t_g}$, excludes auxiliary labels from endpoint lists, and multiplies by

$$K_\nu D^{i\zeta} q^{(\omega(n) - Ms)/2} q^{(\omega(n') - Ms)/2}. \tag{2.2}$$

Sum the patterns and apply the orthogonal permutation averages on both sides. This is $\mathcal{A}$ on $\mathcal{H}_{\mathrm{ph}}$, in the source's row convention. The ideal operator $\mathcal{T}(\Theta)$ samples unshared big labels independently and has multiplier $K_\nu D_{\mathcal{B}}^{i\zeta} e^{2\pi i \Theta D_{\mathcal{B}}}$. Its Hilbert space uses product *probability*, not physical state mass.

The joint physical measure per group is $V_g^{-M-t_g}$ in either direction. Thus the adjoint reverses kD, swaps lists, and conjugates the multiplier; it is not in general $\mathcal{A}$. The source's Schur estimate follows from

$$\sup_{z \geq 0} V_g^{-t_g} (z + t_g)_{t_g} q^{z/2} < \infty, \qquad t_g \leq \ell + m,$$

uniformly in J, and the pattern budget $\sum_\nu \sup |K_\nu| \leq L^{C_1}$. This gives an absolute bound $\|\mathcal{A}\| \leq L^{C_{\mathrm{abs}}}$, not cancellation.

2.1 Finite Ram energy without a symmetry error

For finite input/output state sets, let a_{st} be the actual row coefficients after all weights and symmetrizations have been summed. If their positive diagonal mass matrices are D_s, D_t, the Euclidean matrix is

$$B = D_s^{1/2} a D_t^{-1/2}, \qquad \langle f, ag \rangle_{\mu_s} = (D_s^{1/2} f)^* B (D_t^{1/2} g). \tag{2.3}$$

Consequently B^*B, BB^* and

$$\mathscr{D}_B = \begin{pmatrix} 0 & B \\ B^* & 0 \end{pmatrix}$$

are legitimate positive-energy or self-adjoint spectral objects. The nonzero eigenvalues of $\mathscr{D}_B$ are $\pm$ the singular values of B. Its functional calculus is the precise finite extension of the spectral route in TR190 needed here. There is no assertion that a is a real symmetric graph adjacency or a positive

Laplacian. A nilpotent matrix $\left(\begin{smallmatrix} 0 & K \\ 0 & 0 \end{smallmatrix}\right)$ has only zero eigenvalues and norm $|K|$, demonstrating the danger of that substitution.

Let u_n be one on all states at n. Then

$$\|u_n\|^2 = \prod_g V_g^{-M}(\omega_g(n))_M. \tag{2.4}$$

These vectors are orthogonal across positions, but not unit vectors or equally weighted vectors. The moment is $\langle u_n, (\mathcal{A}\mathcal{A}^*)^R u_n\rangle$; its paths return to the integer position, not necessarily to the same list. Replacing it by a trace over individual lists changes the problem.

For a finite calculation let H bound every step $|kD|$ and include all states at positions within $2RH$ of the tested positions. Every contributing length-$2R$ path stays in this enlargement. Compressing $\mathcal{A}$ there therefore reproduces the tested moments exactly; an arbitrary small box does not. This is the applicable polynomial-locality statement. It does not provide a practical complexity bound for enumerating all states.

3 Frequency and logarithmic representations

For an alternating path of length $N = 2R$, after residue replacement the source uses

$$\mathbf{1}_{h_N=0} = \int_0^1 e^{2\pi i\theta h_N}\, d\theta.$$

Small-prime residues have Haar probability. Pending residues have *counting* measure. On $\mathbb{Z}/p\mathbb{Z}$ the unitary transforms are respectively $p^{-1}\sum_r f(r)e^{2\pi i - jr/p}$ and $p^{-1/2}\sum_r f(r)e^{2\pi i - jr/p}$, with counting measure on the frequency side in both cases. Translation by Δ multiplies these coefficients by $e^{2\pi i j\Delta/p}$ for the convention $f(r) \mapsto f(r+\Delta)$. Tensorization is finite and exact. Fixing the small shared product and the pending prime lists leaves $\mathcal{T}(\Theta)$ or its adjoint on the big active lists. The modulus-one scalar is harmless; the changed active density and forbidden free/active coincidences are separate perturbations in source `02`, lines 986–1018.

This is an additive-residue Fourier decomposition, not a Mellin diagonalization of divisibility. In particular $p \mid n$ and $p \mid n + kp$ are the same condition and cost $1/p$, not $1/p^2$. The source memory space

$$\bigoplus_{j\ge 0} L^2_{\mathrm{sym}}\left(\mathcal{Z}_g^j, \lambda_g^{\otimes j}/j!\right), \qquad d\lambda_g(p,r) = \frac{d\,\mathrm{count}(r)}{V_g p b_p}, \quad b_p = 1 - p^{-1}\sum_{i=0}^{N}(1-q_i), \tag{3.1}$$

uses $q_0 = q_N = \sqrt{q}$ and $q_i = q$ for $0 < i < N$. It pays the reciprocal-prime factor at birth. Later residue-zero tests cost one point of counting measure. The factorial measure is in the inner product and adjoint, not inserted at each row transition.

For the separate smooth scale variable, the actually read private proof gives, for $g_c(t) = e^{ct}f(e^t) \in L^1 \cap L^2$,

$$\mathcal{M}f(c+i\zeta) = \widehat{g}_c(-\zeta), \qquad \int_0^\infty |f(y)|^2 y^{2c-1}\, dy = \frac{1}{2\pi}\int_{\mathbb{R}} |\mathcal{M}f(c+i\zeta)|^2\, d\zeta, \tag{3.2}$$

where $\widehat{g}(\zeta) = \int g(t)e^{-it\zeta}\, dt$. Thus dy/y corresponds to $c = 0$, ordinary dy to $c = 1/2$. For $\Phi(t) = \Psi(e^t)$ the source lift has exactly

$$\Psi(n/(ZD)) = \frac{1}{2\pi}\int \widehat{\Phi}(v)(n/Z)^{iv}D^{-iv}\, dv.$$

The $1/n$ in the lifted correlation is retained. In source `04`, lines 421–536, $h = H/Y$ and $dv \asymp Y\, du$ give the factor $H^{-2}Yh^2 = Y^{-1}$ in the local Mellin-energy bound. That bound still requires estimates for the resulting arithmetic Dirichlet polynomials; Plancherel does not supply their cancellation.

4 A spectrally truncated replacement for ideal cells

This section proves the principal local improvement. It concerns the comparison lemma in source 03, not the subsequent prime-extraction theorem. All spaces below are finite probability spaces, so no limiting interchange is involved. The inner product is conjugate-linear in its first argument.

Let Ω_Λ be the full ordered tuple space for a nonempty set of selected slots, at most m per big group. Its product is $X > 0$. Partition by $\log X \in [jh, (j+1)h)$, $0 < h \le 1$, with masses ν_j; retain cells with $\nu_j \ge \xi > 0$. Let $\chi_1, \dots, \chi_{n_Q}$ be all primitive characters of conductor at most Q, including conductor one. Here $n_Q \le Q^2$. Assume the selected primes exceed Q^2, so all these characters have modulus one on the products. No character equidistribution is assumed. On retained cell j define

$$(C_j c)(X) = \sum_{a=1}^{n_Q} \chi_a(X) c_a, \qquad G_j = C_j^* C_j, \qquad (G_j)_{ab} = \mathbb{E}[\overline{\chi_a(X)} \chi_b(X) \mid j].$$

For $0 < \tau \le 1$ use the spectral inverse

$$G_{j,\tau}^\dagger = \sum_{\lambda \ge \tau} \lambda^{-1} P_{j,\lambda}.$$

Eigenvalues below τ, including zero, contribute zero. Set

$$\mathcal{K}_{\Lambda,\tau}(D, X) = \sum_{j : \nu_j \ge \xi} \frac{\mathbf{1}_j(D) \mathbf{1}_j(X)}{\nu_j} \sum_{a,b=1}^{n_Q} \chi_a(X) (G_{j,\tau}^\dagger)_{ab} \overline{\chi_b(D)}. \tag{4.1}$$

The orientation is deliberate: $\Pi_\Lambda f(X) = \mathbb{E}_D \mathcal{K}_{\Lambda,\tau}(D, X) f(D)$. It agrees with the source kernel when the inverse matrix is replaced by the identity. Here cross-character terms, not only diagonal character pairs, are allowed.

Theorem 4.1 (Cell projection and character residual)**.** *On each retained conditional space, $C_j G_{j,\tau}^\dagger C_j^*$ is an orthogonal projection. The global Π_Λ is an orthogonal projection, zero on discarded cells, and*

$$\|\Pi_\Lambda\| \le 1, \qquad \sup_{D,X} |\mathcal{K}_{\Lambda,\tau}(D, X)| \le \frac{n_Q}{\xi \tau}, \tag{4.2}$$

$$\|(1 - \Pi_j)\chi_a\|_{L^2(\Pr(\cdot \mid j))}^2 \le \tau. \tag{4.3}$$

These conclusions allow singular Gram matrices and arbitrarily correlated characters.

Proof. For a positive eigenpair $G_j v = \lambda v$, $\|v\| = 1$, the vectors $C_j v / \sqrt{\lambda}$ are orthonormal. The displayed operator is their orthogonal projection for $\lambda \ge \tau$. Different cells are orthogonal summands. For the kernel bound, both evaluation vectors have Euclidean norm $\sqrt{n_Q}$, and $\|G_{j,\tau}^\dagger\| \le 1/\tau$. Finally $\chi_a = C_j e_a$, and its discarded squared norm is

$$e_a^* \left(\sum_{0 < \lambda < \tau} \lambda P_{j,\lambda} \right) e_a \le \tau.$$

Zero eigenvalues have zero image under C_j. This proves all assertions. □

The source uses Siegel–Walfisz to make G_j close to the identity and then uses the uncorrected frame operator $C_j C_j^*$, of norm at most two. Theorem 4.1 replaces that step. Exact pseudoinversion without truncation would reproduce all characters exactly, but could have uncontrolled coefficients. Truncation bounds those coefficients and quantifies precisely what is lost; it is not an assertion of exact reproduction for the discarded spectral directions.

4.1 Major-arc comparison in the product probability space

Take independent tuple spaces A, B. For any tuple tests f on B and g on A, the relevant multiplier is

$$F(X_A, X_B) = (X_A X_B)^{i\zeta} e^{2\pi i \alpha X_A X_B}.$$

The comparison multiplier is $(\Pi_A \otimes \Pi_B)F$. The tests need not depend only on the products. Their product has L^2 norm $\|f\|_2\|g\|_2$ by independence.

Proposition 4.2 (Explicit major-arc error)**.** *Restrict the tests to $X_A \in [U, 2U)$ and $X_B \in [V, 2V)$, $U, V \geq 1$. Suppose $\alpha = u/r + \beta$, $(u, r) = 1$, $r \leq Q$, and all selected primes exceed Q^2. Let ϵ_A, ϵ_B be the discarded-cell probabilities. Put*

$$D_M = |\zeta| + 8\pi e^2 |\beta| UV.$$

Then the absolute comparison pairing error is at most

$$\left(\sqrt{\epsilon_A + \epsilon_B} + 4hD_M + 2\sqrt{Q\tau}\right) \|f\|_2\|g\|_2. \tag{4.4}$$

The same kernels work for every α, ζ.

Proof. The comparison is zero if either tested product is in a discarded cell. There $|F| = 1$, and Cauchy–Schwarz bounds the raw pairing by the first term of (4.4). This does not claim that restriction to a small event has small operator norm.

Consider retained cell pairs intersecting the two test dyads. Every product of labels in these cells is at most $4e^{2h}UV$; hence the derivative of $M(v) = e^{i\zeta v} e^{2\pi i \beta e^v}$ is bounded by D_M. Choose the sum of the two log-cell centers as v_0 for each pair. The change from $M(\log X_A + \log X_B)$ to $M(v_0)$ has magnitude at most $2hD_M$. It costs this in L^2 before projection and no more after projection, giving $4hD_M$. All projections preserve their cells, so restricting attention to this union of cell pairs is legitimate.

On units modulo r, finite Fourier expansion gives

$$e^{2\pi i uv/r} = \sum_{\psi \bmod r} c_\psi \psi(v), \qquad \sum_{\psi} |c_\psi| \leq \sqrt{\varphi(r)} \leq \sqrt{Q}.$$

Each ψ agrees on these products with its primitive inducing character in the chosen family. On a conditional cell pair, the contraction property and (4.3) give

$$\|\psi \otimes \psi - (\Pi_{Aj}\psi) \otimes (\Pi_{Bk}\psi)\|_2 \leq 2\sqrt{\tau}.$$

The scalar $M(v_0)$ has modulus one. Sum squared norms over disjoint cell pairs with their probability masses, then sum the character coefficients by the triangle inequality. This costs $2\sqrt{Q\tau}$, not a cell-count factor. Pair the resulting L^2 residual with $\overline{f} \otimes g$ to finish. □

Compared with source `03`, lines 372–408, this replaces its $O(Q^4 h(S + L^{C_4}))$ pointwise log error by an $O(h(S + L^{C_4}))$ pairing error, and replaces its character equidistribution error by $2\sqrt{Q\tau}$. This uses Hilbert-space contraction, not a pointwise claim about an inverse Gram kernel's absolute row sum. The price is the factor τ^{-1} in coefficient complexity.

4.2 Minor arcs, arbitrary prime subsets, and choice order

Unique factorization and the harmonic slot laws give the source's exact product atom, for $k_g \le m$ selected slots and multiplicities a_p,

$$\Pr(P_\Lambda = n) = \frac{1}{n} \prod_g \frac{k_g!}{V_g^{k_g} \prod_{p \in \mathcal{P}_g} a_p!} \le L^\kappa / n, \tag{4.5}$$

with $\kappa = C_0[\log(m!) + m \log \max(1, v_-^{-1})]$. For an arbitrary ordered tuple test f, putting $a_n = \mathbb{E}[f \mathbf{1}_{P_\Lambda = n}]$ gives $\sum |a_n|^2 \le L^\kappa (cU)^{-1} \|f\|_2^2$ on $[cU, CU]$. This is a fiber Cauchy–Schwarz bound, not an assumption that the test already factors through the product.

For completeness the needed integer estimate follows by expanding after Cauchy–Schwarz, obtaining

$$\|a\|_2^2 \|b\|_2^2 \left[U + \sum_{1 \le h' \le CV} \min(U, \|\alpha h'\|_{\mathbb{R}/\mathbb{Z}}^{-1}) \right].$$

If $\alpha = u/r + \beta$, $|\beta| \le r^{-2}$, split the h' range into blocks of diameter at most $r/2$. Different phases within a block are $1/(2r)$-separated, so its sum is $O(U + r \log(2r))$. For $r = 1$ use the trivial bound. Combining with (4.5) yields, on product dyads, the bilinear norm bound

$$CL^\kappa \left(V^{-1} + r^{-1} + \frac{\log(2r)}{U} + \frac{r \log(2r)}{UV} \right)^{1/2}. \tag{4.6}$$

The Mellin factor separates into two unit-modulus factors for every real ζ. This proof requires no prime-interval or prime-distribution hypothesis. It is the elementary estimate read in `03`, lines 110–174, restated here rather than credited as new.

In the comparison term, move the two projection kernels onto the tests. The resulting operators, their adjoints and complex conjugates have norm at most one. The transformed tests stay in $[U/e, 2eU]$ and $[V/e, 2eV]$. Thus (4.6) also bounds the comparison term, with an absolute change in constants. Dirichlet approximation with denominator at most UV/S gives

$$1 \le r \le UV/S, \quad |\beta| \le r^{-2}, \quad |\beta| UV \le 2S/r.$$

For $r > Q$ the error is $O(L^\kappa [Q^{-1} + L^{d+1}/S + o(L^{-A})]^{1/2})$. For $r \le Q$ use Proposition 4.2 and $\epsilon_A + \epsilon_B \le (N_A + N_B)\xi$, where $N_\Lambda \ll_m 1 + TL^d/h$.

Theorem 4.3 (Weaker-assumption comparison)**.** *Fix an integer $m \ge 1$, $A_0 > 0$, $C_4 \ge 0$ and the source's big-group range, disjointness and harmonic-mass bounds. The big groups may be arbitrary finite subsets of primes in $[\exp(L^c), \exp(L^d)]$; they need not be intervals. There are fixed powers $Q, h^{-1}, \xi^{-1}, \tau^{-1}$ of L, independent of J, such that the kernels (4.1) have comparison error at most*

$$L^{-A_0} \|f\|_2 \|g\|_2$$

for every real α, every $|\zeta| \le L^{C_4}$, and all ordered tuple tests on nonempty selected slot sets of at most m slots per group, once x is sufficiently large. They have polynomial coefficient and finite-expansion complexity, independent of J. No Siegel–Walfisz estimate or Gram conditioning hypothesis is used in this theorem.

Proof. There are at most L^{d+1} product dyads for each selected set, and each product exceeds $\exp(L^c)$. Choose the following fixed exponents in order:

$$\begin{aligned} B_0 &> A_0 + 2(d+1) + 10, \\ q_*, s_* &> 2(B_0 + \kappa + 10) + d + 1, \\ h_* &> \max(s_*, C_4) + B_0 + 10, \\ t_* &> 2B_0 + q_* + 10, \\ x_* &> 2B_0 + d + 1 + h_* + 10. \end{aligned}$$

Set $Q = L^{q_*}$, $S = L^{s_*}$, $h = L^{-h_*}$, $\tau = L^{-t_*}$, $\xi = L^{-x_*}$. The preceding minor and major estimates give $O(L^{-B_0})$ per dyad pair, with constants absorbed by the margins. Summing proves the assertion. The selected primes eventually exceed Q^2. There are polynomially many cells and at most Q^2 characters; each cell matrix has at most Q^4 entries, each bounded in magnitude by τ^{-1}. Equation (4.2) gives $\sup|\mathcal{K}| \le Q^2\xi^{-1}\tau^{-1}$. These choices do not involve J. The conditional Gram matrices may depend on x and the selected groups; that is permitted for the source's kernel family. □

This is an existence and finite representation theorem, not a claim that these exact conditional Grams are inexpensive to compute. Truncation is discontinuous at an eigenvalue equal to τ; numerical use needs a certified gap or an explicitly budgeted residual. Arithmetic distribution is still needed elsewhere in the manuscript. In particular this theorem does not remove interval assumptions from its shifted-correlation or prime-mass arguments.

5 Sharper symmetric-probe accounting

The source decomposes symmetric endpoint vectors after averaging the last ℓ slots. For $M = J + \ell$, its exact surviving fraction on a component of degree j in one group is

$$a_{M,m,\ell}(j) = \frac{\binom{M-m-\ell}{j-m}}{\binom{M}{j}} = \frac{(j)_m (M-j)_\ell}{(M)_{m+\ell}}, \tag{5.1}$$

with value zero for impossible cardinalities. This formula is source algebra, not an M021 invention. Keeping its maximum instead of replacing it by $2\ell^\ell m^{-\ell}$ gives a finite, computable sharper constant.

Lemma 5.1 (Exact degree location)**.** *For $m, \ell \ge 1$ and $M \ge m + \ell$, put $t = m(M+1)/(m+\ell)$. A maximizer of (5.1) is $\lfloor t \rfloor$, clipped to $[m, M-\ell]$. If t is integral, the adjacent index $t-1$ also maximizes when admissible. Define the maximum to be $\rho_{M,m,\ell}$.*

Proof. For consecutive admissible degrees, the ratio is

$$\frac{a(j+1)}{a(j)} = \frac{j+1}{j+1-m} \frac{M-j-\ell}{M-j}.$$

It is at least one exactly when $m(M-j) \ge \ell(j+1)$, equivalently $j \le t-1$. This proves the stated location and tie convention. □

The fraction tensorizes across groups even for non-product tests, by orthogonality and within-group symmetry. With s_i groups in block i, the full-probe norms are at most $\rho_{M,m,\ell}^{s_i/2}$ times the original norm. No large-M asymptotic is needed for this finite bound.

Proposition 5.2 (Orthogonal error sum)**.** *Suppose a comparison for each admissible component/free-subset pair has error at most $\epsilon\|F_S\|_2\|H_T\|_2$. If the two blocks have P_1, P_2 probes, the total error is at most*

$$\epsilon\sqrt{b(P_1)b(P_2)}\|F\|_2\|H\|_2, \qquad b(p) = 5^p - 2\cdot 3^p + 2^p. \tag{5.2}$$

For $P = P_1 + P_2 > 0$ this is at most $\epsilon 5^{P/2}\|F\|_2\|H\|_2$, improving the source's $\epsilon 3^P$ estimate.

Proof. For a component $S \subseteq \mathcal{I}_1$ there are $2^{P_1-|S|} - 1$ nonempty free subsets of its missing probes, and similarly in the other block. Sum the error bounds before replacing component norms. Cauchy–Schwarz and orthogonality give

$$\sum_S (2^{p-|S|} - 1)\|F_S\| \le \left[\sum_S (2^{p-|S|} - 1)^2\right]^{1/2} \|F\|.$$

The bracket is $5^p - 2\cdot 3^p + 2^p$ by the binomial theorem. Multiply the two block estimates. No independence of the component errors or additional cancellation was assumed. □

Corollary 5.3 (Residual ideal operator)**.** *Use the source's asymmetric coefficients*

$$-(-1)^{|A|+|B|}\mathcal{K}_A(D_A, X_A)\mathcal{K}_B(D_B, X_B),$$

with the new kernels. For symmetric tests its norm is at most

$$\rho_{M,m,\ell}^{s_1/2} + \rho_{M,m,\ell}^{s_2/2} + L^{-A_0}\sqrt{b(P_1)b(P_2)}. \tag{5.3}$$

In particular the source's arbitrary fixed ideal accuracy can be attained for arbitrary big-prime subsets satisfying the preceding range/mass hypotheses, with polynomial complexity independent of J.

Proof. If a freed first-block coordinate occurs in the input mean-zero component, the term vanishes; the multiplier omits that input coordinate. The corresponding second-block statement concerns the target coordinate. For surviving terms condition on all other shared labels; their product only changes the real additive frequency and a unit-modulus scalar. Theorem 4.3 applies to the remaining arbitrary tuple tests. For each pair with nonempty missing sets the alternating nonempty subset sums both equal -1, so their product cancels the raw term. The uncancelled terms are exactly

$$R(F_{\text{full}}, H') + R(F' - F_{\text{full}}, H_{\text{full}}).$$

The raw multiplier has modulus one, so their total is bounded by the first two terms in (5.3). Proposition 5.2 handles the approximation errors. This repeats the source's cancellation mechanism with the proved replacement and sharper accounting.

For a target $L^{-E_{\text{id}}}$, first choose m so the source's upper bound $\theta_m = 2\ell^\ell m^{-\ell}$ obeys $-(c_0/6)\log\theta_m > E_{\text{id}} + 3$. Choose $A_0 > E_{\text{id}} + \frac{1}{2}mC_0\log 5 + 10$, then the comparison precisions. There are at most $L^{mC_0\log 2}$ patterns. A sufficient coefficient budget is any

$$C_1 > mC_0\log 2 + 4q_* + 2x_* + 2t_* + 1.$$

Only then choose $J \ge m$ and $J + \ell \ge (m+\ell)(m+\ell-1)$. The original bound on θ_m ensures the asserted asymptotic target; the exact finite ρ can improve individual choices. All kernel parameters precede J. □

This changes the ideal construction only. Invoking physical transference still requires its residue replacement, full memory identity, rare-transfer and equality-rank estimates. Reading their text does not make a separate global proof audit automatic. Nor does the replacement automatically propagate through every downstream separation formula for the now cross-character coefficients.

6 Memory cutoff and endpoint assumptions

6.1 A smaller sufficient memory budget

The source takes $N = 2\lceil L^{0.52}\rceil$ and total memory cutoff $B = \lceil L^2\rceil$. A path can receive at most $S_0 = (\ell+m)|\mathcal{B}|N$ particles from stores. If it ever exceeds memory B, its total ghost births G exceed $B - S_0$. The doubled-birth absolute path estimate in `02`, lines 681–822, is precisely the input to the next elementary improvement.

Proposition 6.1 (Explicit memory tail)**.** *Let nonnegative absolute path weights $w(\pi)$ satisfy $\sum_\pi 2^{G(\pi)}w(\pi) \le KL^{C(N+1)}$, with $K \ge 1$ and $C \ge 0$ fixed. Then*

$$\sum_{\pi:\max\text{ memory}>B} w(\pi) \le K2^{S_0-B}L^{C(N+1)}. \tag{6.1}$$

For desired L^{-AN} error it suffices to take

$$B \ge S_0 + \frac{[AN + C(N+1)]\log L + \log K}{\log 2}. \tag{6.2}$$

For every fixed A simultaneously, $B = \lceil N(\log L)^2\rceil$ gives $o(L^{-AN})$ as $L \to \infty$.

Proof. On the omitted event $1 \le 2^{G-(B-S_0)}$. Sum this inequality against w. For the last assertion, $S_0 = O(N\log L)$, whereas $B/(N\log L) \to \infty$. Taking logarithms in (6.1) proves it. □

This is a proved improvement of the truncation deduction *conditional on the stated absolute generating bound*, not a fresh proof of that arithmetic-memory bound. It is uniform in paths, and keeps all factorial and residue conventions. It does not assert that the smaller memory space is computationally small.

In the source rare-transfer estimate $2L^{C_a}\sqrt{B/\mathcal{D}}$, $\mathcal{D} = \binom{J+\ell}{\ell}^{|\mathcal{S}|}$, the universal new choice has

$$\sqrt{B} = O(L^{0.26}\log L),$$

instead of $O(L)$. Thus a sufficient asymptotic inequality is $\frac{c_0}{2}\log\binom{J+\ell}{\ell} > C_a + G + 0.26$ with a fixed positive margin to absorb $\log L$ and constants. The ideal accuracy still precedes J. The birth-address count $M|\mathcal{B}|(N+1) + (N+1)|\mathcal{B}|B$ remains polynomial in L; the rank argument still needs its own signed estimates. Only this cutoff and its explicit downstream size costs are being improved.

6.2 Pairings controlled by local coefficient energy

Let a bounded operator A on a weighted discrete state space have position propagation at most $H \ge 1$. Let u_n be the unnormalized fiber-constant vectors, and $f = \sum_{n\in I} f_n u_n$. Partition I into blocks I_j of length $\lceil H\rceil$. Set

$$s_j = \sum_{n\in I_j} |f_n|^2, \quad d_j = \sum_{n\in I_j} \langle u_n, (AA^*)^R u_n\rangle, \quad s_* = \max_j s_j.$$

Notice that s_j is coefficient energy, not the physical norm of f_j.

Proposition 6.2 (Localized endpoint bound)**.** *For every integer $R \geq 1$ and every g, if the H-enlarged blocks have overlap multiplicity at most Ω, then*

$$|\langle f, Ag\rangle| \leq \sqrt{\Omega}\|g\|\|f\|^{1-1/R}\left(\sum_j s_j d_j\right)^{1/(2R)}. \tag{6.3}$$

For $R = 1$ the factor $\|f\|^0$ is interpreted as one; zero vectors are treated directly. In particular one may replace the last sum by $s_* \sum_j d_j$.

Proof. Restrict g to each enlarged block, obtaining g_j with $\sum\|g_j\|^2 \leq \Omega\|g\|^2$. Spectral Hölder for AA^* gives

$$|\langle f_j, Ag_j\rangle| \leq \|g_j\|\|f_j\|^{1-1/R}\langle f_j, (AA^*)^R f_j\rangle^{1/(2R)}.$$

The coefficient matrix $[\langle u_n, (AA^*)^R u_m\rangle]_{n,m\in I_j}$ is positive semidefinite. Its largest eigenvalue is bounded by its trace d_j, even though the u_n are not normalized. Hence the last quadratic form is at most $s_j d_j$. Sum by Hölder with exponents 2, $2R/(R-1)$ and $2R$. For $R = 1$ use Cauchy–Schwarz on the remaining two factors. □

If the source root moment satisfies $\sum_j d_j \leq |I \cap \mathbb{Z}|L^{-2RE}$, its pairing conclusion follows under the weaker assumption $\log\max(1, s_*) = O(R)$, with the same endpoint norm bounds. The explicit loss is $s_*^{1/(2R)}$. The source's $\sup|f_n| \leq \exp(O(\sqrt{L}))$ implies this because $\log H = O_J(TL^d) = o(R)$. The local energy formulation also permits larger sparse spikes, and (6.3) is sharper if they occur in blocks with smaller moment mass. It does not allow arbitrary mark-dependent first endpoints: those would need a different root test.

7 Finite certificates and their failure modes

The projection construction can be checked on finite conditional measures without asserting arithmetic cancellation. After whitening, a feature matrix C has $G = C^*C$. A verified spectral split gives $P = CG_\tau^\dagger C^*$, and direct identities test $P = P^* = P^2$ and the character residuals. Floating-point eigenvalues alone do not prove which side of τ a nearly equal eigenvalue lies on.

For approximate finite bilinear calculations, if $\|B - \widetilde{B}\| \leq \eta$, $\|x - \widetilde{x}\| \leq \epsilon_x$ and $\|y - \widetilde{y}\| \leq \epsilon_y$, then, by adding and subtracting,

$$|x^*By - \widetilde{x}^*\widetilde{B}\widetilde{y}| \leq \eta\|x\|\|y\| + \|\widetilde{B}\|(\epsilon_x\|y\| + \|\widetilde{x}\|\epsilon_y). \tag{7.1}$$

This applies to the actual weighted matrix (2.3); an entrywise bound can supply η through the Frobenius norm. High moments are less stable: for $S = BB^*$ and $\widetilde{S} = \widetilde{B}\widetilde{B}^*$,

$$\|S^R - \widetilde{S}^R\| \leq R\max(\|S\|, \|\widetilde{S}\|)^{R-1}\|S - \widetilde{S}\|, \quad \|S - \widetilde{S}\| \leq (\|B\| + \|\widetilde{B}\|)\eta.$$

Both follow from telescoping products. Thus successful small finite tests cannot certify $R \asymp L^{0.52}$ asymptotic cancellation without uniform error budgets.

Nor may one impose global distinct births independently at each edge. The source's partition inclusion–exclusion uses equality rank r, with at most $2r$ affected births. High rank pays repeated fresh point-mass costs; low rank retains signed contraction on most edges. A Parseval change of coordinates preserves these difficulties rather than solving them. The manufactured repeated-prime sanity checks communicated by the parent concern normalization only; they are not source arithmetic proofs.

8 Arithmetic obligations and the totient boundary

The report's local ideal result leaves the following substantive tasks.

1. The physical transference theorem requires uniform CRT errors after absolute path summation, the exact lifespan identity, factorial adjoints, rare transfers, and equality-rank control. The full source text was read, but the entire theorem is not independently certified by this report.

2. The shifted endpoints must retain the form

$$F_0(n) = \sum_{mpr=n_*} \alpha_m \chi_0(m) m^{i\sigma_0} \mathbf{1}_{p\in I_p,\ P^-(p)>W} p^{i\sigma_1} c_r,$$

   where this p is a rough integer, not necessarily prime. The first coefficient requires small harmonic character/Mellin sums on every specified interval. The local Mellin energy does not prove the exceptional time or long-rough-factor estimates.

3. Type II must convert $mn = 2u + 1$ without changing the discrepancy-bearing sequence. The read determinant calculation gives $m' - m = 2ek$, $h' - h = kn$, $z = m'h$, $z + k = mh'$. Its converse needs $e = (m' - m)/(2k) > 0$ integral and $(m, k) = 1$; otherwise divisibility is not recovered. The smooth split, diagonal saving and uniform separation costs are additional obligations, not supplied by the finite determinant identity.

4. Candidate mass, Type I distribution, proxy discrepancy, the density integral, and balanced-composite estimates must all hold with the stated noncircular order of parameters. In particular the balanced-range constant must precede the choice of its small gap. These proofs were not fully read or checked here.

The final source deduction can be understood conditionally. Given more than $X^{1-\delta}$ primes $p \le X$ with $p - 1$ X^δ-smooth, take $k = \lfloor X^{2\delta} \rfloor$, $3\delta < 1$. There are more than $X^{(1-3\delta)k}$ distinct squarefree products of k such primes. Their totients are at most X^k and have at most

$$(1 + k \log X/\log 2)^{X^\delta} = X^{o(k)}$$

possible values. Pigeonholing gives a fiber at least $X^{(1-3\delta-o(1))k}$. Taking $4\delta < \varepsilon < 1$ yields the claimed exponent relative to $n \le X^k$. Each fiber is finite since $p^{a-1}(p-1) \mid \varphi(m)$ when $p^a \parallel m$, so growing fiber lower bounds force infinitely many distinct values. This is the source's classical conditional extraction, not a new M021 result. Its prime-supply premise remains an obligation. No uniform $\delta \downarrow 0$ claim is made.

9 Public comparison, costs, and research priorities

Public-title searches and primary abstract/version-page reads were made on October 8, 2026. No private text or private titles were submitted. Lichtman's primary record [2] gives the fixed smoothness exponent 0.2844 and quadrilinear moduli up to $x^{17/32}$. This is a different level of arithmetic assertion from a finite projection identity. Helfgott–Radziwiłł [3] obtain norm bounds for a centered, essentially symmetric divisibility graph on a large subset. Their graph does not justify treating the present complex weighted operator as self-adjoint. Pilatte's version 3, dated August 25, 2026 [4], gives a power saving in the logarithm for the logarithmically weighted two-point Liouville sum. These primary abstracts were read, not their full proofs; no new proof of those results is claimed.

Bharadwaj–Rodgers, version 4 of April 9, 2026 [5], distinguish level-one distribution, which gives the full Poisson–Dirichlet law under their hypotheses, from positive distribution level and restricted-support correlation results. This distinction reinforces the report's boundary: representing a finite energy does not establish a distribution hypothesis. The public repository identifies the assigned preprint, but publisher provenance does not certify its headline conclusions. This bounded refresh is not an exhaustive literature or historical-priority search.

Advantages. The new ideal comparison works with arbitrarily correlated conditional characters and arbitrary prime subsets in the specified ranges. It gives explicit coefficient costs, norm-one projections, and an error budget whose choice order is independent of J. The probe and memory refinements reduce concrete losses rather than rename existing coordinates. The endpoint estimate identifies the exact local coefficient norm actually used.

Costs and limitations. The kernels require finite conditional Gram data and cross-character coefficients; they need not be positive. The factor τ^{-1} can make the pointwise complexity large, so the improved log-cell accuracy is not automatically an implementation speedup. There is no lower Gram eigenvalue assumption, but numerical truncation still needs certification. Mark length and all other constants remain fixed before x grows. The finite-state representation can be enormous, and no uniform arithmetic norm follows from its positivity.

Next proofs. The first concrete follow-up is to propagate the cross-character expansion through every shifted-correlation separation step, retaining the exact discrepancy modulus and frequency budgets. Next, rigorously enclose selected conditional Grams and quantify the practical coefficient cost against the source's uncorrected frames. Finally, audit the memory equality-rank estimates and the prime-extraction uniformity independently. These are research tasks, not completed advances credited to the present score.

10 Verification and contribution assessment

The accompanying bounded tests exercise weighted adjoints and singular energies, repeated-prime normalization, singular and correlated character Grams, projection residuals, a finite major-arc comparison, exact probe maxima/error counts, memory budgets, endpoint inequalities and determinant reconstruction. Synthetic matrices and finite prime sets test formulas; the recorded total is 7,649 cases: 6,863 exact and 786 numerical. They are not an implementation of the full source transference or sieve. Floating-point checks are diagnostics, not interval certificates or proofs. The report contains the local proofs independently of those tests.

Native opening and compilation are attempted first; an unchanged existing MiKTeX helper supplies the exported PDF when the native compiler is unavailable. The audit records the actual outcome, all-page visual review, read intervals, hashes, test counts, and the frozen file inventory. No shared queue, catalogue, helper, original source, or other report is edited.

The demonstrated contribution is **C3/5**, independent of L3. The principal evidence is Theorem 4.3 and its proved ideal-operator consequence, which weaken a concrete comparison hypothesis; Proposition 5.2 and the memory/endpoint propositions are smaller quantitative refinements. Classical spectral calculus, Cauchy–Schwarz, the source's elementary bilinear estimate and its totient pigeonhole argument are not credited as inventions. No historical priority, global arithmetic breakthrough, smooth-prime theorem, or totient conjecture resolution is certified here.

References

[1] OpenAI, *Weighted dilation graphs, smooth shifted primes and totient fibers*, September 24, 2026, pinned multi-file build. Public identification record.

[2] J. D. Lichtman, *Primes in arithmetic progressions to large moduli, and shifted primes without large prime factors*, 2022. https://arxiv.org/abs/2211.09641.

[3] H. A. Helfgott and M. Radziwiłł, *Expansion, divisibility and parity*, version 2, 2021. https://arxiv.org/abs/2103.06853.

[4] C. Pilatte, *Improved bounds for the two-point logarithmic Chowla conjecture*, version 3, August 25, 2026. https://arxiv.org/abs/2310.19357.

[5] A. Bharadwaj and B. Rodgers, *Large prime factors of well-distributed sequences*, version 4, April 9, 2026. https://arxiv.org/abs/2402.11884.

[6] M. Subbarao, TR190, *The Graph-Global Ram Transform*, selected actual local text, lines 1–318. Confidential source, no public link.

[7] M. Subbarao, *RamCalBooks*, volume 2, chapter 1, Mellin inversion and energy, lines 447–543. Confidential source, no public link.

[8] M. Subbarao, TR1674, revised version 2, selected actual energy and finite Gram text, lines 112–380. Confidential source; its external source-reported RH machinery is not independently audited here.